

European Planning Studies

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ceps20

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To cite this article: Roberta Capello & Silvia Cerisola (2020): Catching-up and regional disparities: a resource-allocation approach, European Planning Studies, DOI: 10.1080/09654313.2020.1823323

To link to this article: https://doi.org/10.1080/09654313.2020.1823323

Published online: 01 Oct 2020.

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EUROPEAN PLANNING STUDIES
https://doi.org/10.1080/09654313.2020.1823323

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Catching-up and regional disparities: a resource-allocation approach

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ABSTRACT

While the allocation of regional assets has been theoretically and empirically analysed in regional growth theories, the subsequent logical step on how such allocation affects regional inequalities has not found direct quantitative validation. This paper tries to fill this gap by presenting an original simulation able to capture the relative weight of each economic resource in the process of decreasing regional disparities. Results witness that not all resources play the same role and that inequalities are better overcome when growth assets fit the regional productive specialization. The former result represents a guide for raising policymakers' awareness on resource priority investments, while the latter proves once again that the local productive vocation is dramatically relevant, and that Cohesion Policy cannot be designed without considering local specificities, in line with the theoretical arguments of the Smart Specialization Strategy.

ARTICLE HISTORY
Received 20 May 2020
Revised 21 August 2020
Accepted 2 September 2020

KEYWORDS
Regional disparities; growth assets; convergence; Theil index

1. Explaining spatial imbalances in the EU: a resource-allocation perspective

A long period of overall significant inter-regional wealth convergence within the EU has been interrupted after 2008, when the decrease in spatial disparities has continued at a much slower pace because of the economic and financial crisis (Farole, Goga, and Ionescu-Heroiu 2018; Camagni et al. 2020). This also led to a worrisome wave of populism and to the success of nationalist political parties all over Europe (Farole, Rodríguez-Pose, and Storper 2011; Rodríguez Pose 2017) and, therefore, made regional imbalances a more and more relevant topic from a political perspective.

Consequently, understanding the mechanisms at the basis of the evolution of spatial disparities is drastically back on the scene in order to design and implement strategies meant at controlling and limiting regional imbalances, taking EU regions back to a solid and continuous convergence path. Exploring the drivers of spatial inequality is thus particularly important in order to provide suitable and informed suggestions meant at raising policymakers' awareness on risks and benefits of different approaches to the reduction of spatial disparities and therefore at guiding the allocation of available public resources.

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In fact, there are basically two dimensions of regional disparities, social and economic, that can be read at individual (Kemeny and Storper 2020) and aggregate (territorial) level (Ezcurra et al. 2005; Ezcurra and Pascual 2007). The present work takes an economic territorial perspective and focuses on wealth (per capita GDP) inequality at the regional (NUTS2) level. Within this framework, there are different possible sources determining the current trend in disparities. One source of inequality is certainly the sectoral composition, with its relative efficiency and reallocation processes that determine the level of productivity (Esteban 2000; Ezcurra et al. 2005; Ezcurra and Pascual 2007) and its growth (Capello and Cerisola 2019a). An additional, and widely recognized, source of regional disparities is the spatially heterogeneous distribution of growth assets, meant as key production factors or resources on which economic development lies. Local growth assets explain in fact the different capacity of each sector (and each firm) to be competitive.

However, not all resources are expected to provide the same positive effects on growth (Petrakos, Kallioras, and Anagnostou 2011; Rodríguez-Pose 2013). Some specific resources (or growth assets) favour the economic performance more than others. Human capital, scale and agglomeration economies, innovation, accessibility and infrastructure and – more recently – institutional quality are interpreted as fundamental for local competitiveness. These processes have been extensively studied, focusing on the relationship between the endowment of the above-mentioned resources and the regional economic performance.

Regional growth models have also been widely exploited to explore inter-regional convergence. The mechanisms behind the relationship between growth and disparities remain however difficult to grasp, since it all depends on the spatial distribution of the resources for growth and on their specific role in the reduction of regional imbalances.

To our knowledge, in fact, not much empirical work exists on the relationship between the distribution of the different growth assets and their relative relevance in determining regional imbalances. This paper tries to fill this gap. The main aim of the present work, indeed, is to study the specific role of the spatial allocation in each growth assets endowment in explaining spatial disparities in the EU. The reason for such interest lies in the need for a clear understanding of 'what resources can be more useful for the development of disadvantaged regions, with the final aim of reducing inequality'. The goal – and the main value added of the present study – is therefore to highlight through an original simulation approach what assets (and where) are more effective in reducing disparities. This is important for supporting policymakers' decisions on investment priorities in less developed regions (LDRs).

The paper starts from discussing the relationship between the endowment of growth assets and regional disparities, focusing on the availability of resources in LDRs with respect to the others (Section 2). It then moves to grasp the effect of the selected resources on regional growth based on a traditional econometric regional growth model. The work subsequently proceeds to the more innovative part, where the simulation procedure is presented in order to highlight the relative importance of each resource in terms of spatial inequalities. The methodology considers that LDRs are not identical and are in fact characterized by different sectoral structures. The possibly different effectiveness of growth assets in general and within specific development/specialization patterns is explored (Section 3). The study then provides the econometric (Section 4) and simulation

(Section 5) results of the traditional regional growth model exploited, taking the sectoral structure of each local economy into account. Finally, conclusions are drawn and presented, along with some policy suggestions on resources allocation under a disparities reduction perspective (Section 6).

2. Growth assets endowment and regional disparities

A long stream of literature exists on inter-regional convergence.[1] Such process has been mainly explored through econometric models to different extents inspired by the impressive work of Barro and Sala-i-Martin (1991, 1992, 1995), who – drawing on neoclassical growth models – empirically investigated the topic looking at how income/production growth is explained by their initial level and interpreting the related coefficient in terms of convergence (if negative). Two concepts of convergence are highlighted within this literature: unconditional, when there is just a negative relationship between the initial value and subsequent growth rates, or conditional, when the effects of parameters that might differ across countries and regions are considered, as well.

As explained, the above-mentioned models are conceptualized to prove the existence of convergence or divergence trends. They show different degrees of sophistication and are sometimes focused on the role of one specific element (be it either migration, knowledge spillovers or EU regional policy), but they are not interested in determining the relative weight of different resources in driving convergence trends and their role in the actual level of disparities, which mainly depends on what happens where (Petrakos, Tsiapa, and Kallioras 2016).

Drawing on the methodological framework presented, convergence processes can be conditional on the presence of specific growth assets that have been widely highlighted within the existing theoretical and empirical literature as triggers and facilitators of regional economic performance (and therefore of the catching-up process of LDRs).

In particular, human capital (Lucas 1988; Dakhli and de Clercq 2004; Rodriguez-Pose and Crescenzi 2008) and innovation (Romer 1986; Grossman and Helpman 2001; Lim 2003; Bauer, Schweitzer, and Shane 2012; Stephens, Partridge, and Faggian 2013; Capello and Lenzi 2014) are universally deemed as important factors for regional growth as well as urbanization economies (Alonso 1971; Parr 2002; Rosenthal and Strange 2004) and infrastructure endowment (Aschauer 1989; Barro 1990; Krugman 1991; Andersson and Karlsson 2007). More recently, institutional quality has also gained momentum as an important element in explaining regional economic performance (Rodríguez-Pose 2013; Ezcurra and Rios 2019; Peirò-Palomino, Picazo-Tadeo, and Rios 2020; Rodríguez-Pose and Ketterer 2020; Pumain and Torre 2020).

The present paper aims at building on the solid existing literature on regional growth and convergence, adding one important and original element to the overall picture. The innovative goal is in fact to highlight the relative weight of different growth assets in the reduction of spatial disparities, also considering regional sectoral specificities.

Both the endowment and the efficient exploitation of growth assets, in fact, tend to be concentrated in rich areas, explaining divergence trends through cumulative self-reinforcing processes (Myrdal 1957; Krugman 1991). Therefore, it is reasonable to think that they affect the catching-up process of LDRs towards more advanced ones and so are among the main determinants of regional disparities, although possibly to a different extent.

The endowment of resources and the efficiency in their use are notably two relevant, though separated, issues. Disentangling these two contributions to regional development is not an easy task and their intrinsic interplay is carefully considered in this work (see Section 3).

Among the tangible and intangible resources on which regional growth and therefore the catching-up process depend, in this paper, a particular attention is given to those mentioned among the priorities identified within the Cohesion Policy (EC 2014), by the last Cohesion Report (EC 2017) and within the academic literature (Farole, Rodríguez-Pose, and Storper 2011,2018; Rodríguez-Pose and Ketterer 2020). In more detail, as suggested by the World Bank, the main growth assets regard (Farole, Goga, and Ionescu-Heroiu 2018):

- *innovation*, since regional comparative advantage in the EU is more and more based on the local innovative capacity. Both *radical* and *market innovation* are here taken into account, since they represent two different modes of innovation, the first one being more related to the creation of new knowledge through R&D often performed within large firms and the second one instead associated with the capacity of marketing a product (Capello and Lenzi 2013);
- *human capital*, in terms of education and skills, given its fundamental role in economic growth;
- *accessibility*, since it is believed to be a necessary condition for regional growth; and
- *institutional quality*, since it makes an important difference for the efficiency in the delivery of public policies and, consequently, for economic development. Moreover, it is critical to support regional transformation and to ensure equitable social outcomes.

The reason for this choice relates to the theory but also to the fact that such resources are judged as fundamental local production factors by policymakers at the point that they represent the most relevant shares of funding within the Cohesion Policy. In fact, among the main objectives pointed out for the 2014–2020 programming period, the EU Cohesion Policy aims at strengthening research, technological development and innovation; at promoting sustainable transport and at improving network infrastructure; at investing in education, training and lifelong learning; and at improving institutional quality and efficiency of public administration (EC 2014, 2017).

In order to have an idea of the regional heterogeneity of growth assets endowment, Table 1 presents some descriptive statistics and, especially, displays the results of the ANalyses Of VAriance (ANOVAs) performed to check if the endowment of resources in LDRs – defined as those NUTS2 regions characterized by GDP PPS per capita lower than 75% of the EU average[2] – is significantly different with respect to that in the other regions. The years considered are 2008 and 2012, at the beginning of the crisis period and at the turnover point just after it. The analysis is based on the indicators presented in Table 2.

From Table 1, a huge disparity in the endowment of growth assets emerges, with LDRs being systematically and significantly less endowed with the resources considered, as expected. This creates an uneven geographical distribution of growth assets. Table 1 reports both radical innovation and market innovation. The two indicators are presented for innovation, capturing two different innovation modes. Patents represent a formal

Table 1. Growth assets distribution in LDRs and other EU regions in 2008 and 2012

	2008				2012			
Growth assets	EU	LDRs	Others	ANOVA[§]	EU	LDRs	Others	ANOVA[§]
Human capital (share of graduates over working age pop.)	23.58%	16.64%	25.92%	***	27.32%	19.79%	29.83%	***
Accessibility (Multimodal accessibility per inh.)	0.006	0.003	0.007	**	0.007	0.004	0.007	**
Quality of institutions	0.277	−0.731	0.6333	***	0.283	−0.6411	0.611	***
Radical innovation (patents per thousand inh.)	0.102	0.008	0.134	***	0.084	0.008	0.110	***
Market innovation (trademarks per thousand inh.)	0.115	0.024	0.136	***	0.143	0.035	0.173	***

[§]Significance of ANOVA as follows: ***1%; **5%; *10%.

Table 2. Description of growth assets indicators

Variable	Description	Computation	Year	Source
GDP growth	Economic development	Average annual GDP PPS[a] growth rate over two periods: 2008–2012 and 2012–2016	2008-2012;2012-2016	Eurostat
human capital	Human capital	% of college graduates/labour force	2008; 2012	Eurostat
accessibility	Multimodal accessibility	ESPON multimodal accessibility per million inhabitants	2006; 2012[b]	ESPON, TRACC project
institutions	Quality of Government	University of Gothenburg Quality of Government index	2010; 2013	University of Gothenburg – Quality of Government Institute
radical inn	Radical innovation	No. of patent applications per thousand residents	2008; 2012	Eurostat
mkt inn	Market innovation	No. of trademark applications per thousand residents	2008; 2012	Eurostat

[a]GDP PPS was preferred with respect to the real one since the final objective of the present work is to assess how the selected growth assets affect the catching-up process of LDRs and, subsequently, the level of regional disparities. For measuring spatial imbalances, in fact, the PPS version is more correct. Moreover, PPS data are those taken into account by the European Commission in defining the different categories of regions for the Cohesion Policy. The addition of country fixed effects in the regressions partially controls for the use of GDP PPS, since the deflator is defined at country level. As a control, the same regressions were run using real GDP growth as the dependent variable and the results do not contrast those presented.

[b]The values for 2012 were estimated through linear interpolation exploiting the trend between 2001 and 2006 (available data from the ESPON project).

innovation, radically embedded in R&D activities typical of rich regions. Market innovation, instead, represents the capacity to brand a product or an activity, which does not require R&D, and characterizes a softer innovation (Stoneman 2010). As it is evident from Table 1, low levels of human capital characterize LDRs, and probably their capacity to grow.

As for the other resources, also the level of quality of institutions and accessibility[3] are dramatically lower in LDRs with respect to the others. This is of course of special concern, since corruption and pervasive rent-seeking (weak institutions) may have a negative influence on the provision of public goods and in the design and delivery of appropriate policies, while poor accessibility to markets also further penalizes less advantaged regions.

This overall picture is particularly important to interpret the specific role of different resources in regional growth and their consequent relative effect on the catching-up

process of weaker regions and on the level of spatial imbalances. The quantification of the relative impact of different growth assets on regional disparities is in fact the main value added of the present work and allows us to provide suggestions on public funds allocation, according to a policy-oriented perspective. The methodology is described in the next sections.

3. The relationship between growth assets and regional disparities: methodological approach

3.1. Growth assets and regional development: econometric analysis

In order to investigate the role of different growth assets in regional performance and to study the potentially different effectiveness of the same resources in differently characterized LDRs, the present work proposes a traditional regional growth regression analysis, which is expressed according to two different models.

The first, more general, econometric model is meant to explore the role of the specific regional endowment of the growth assets of interest in the local economic development and is carried out according to the following specification:

$$\begin{aligned}\text{gdp growth}_{i,t,t+4} = {} & \alpha + \beta_1 \text{human capital}_{it} + \beta_2 \text{accessibility}_{it} \\ & + \beta_3 \text{institutions}_{it} + \beta_4 \text{radical inn}_{it} \\ & + \beta_5 \text{mkt inn}_{it} + \beta_6 \text{gdp pc}_{it} + \beta_7 \text{metro}_i \\ & + \beta_8 \text{sh agr}_{it} + \beta_9 \text{sh ind}_{it} + \beta_{10} D + \beta_{11} T + \varepsilon\end{aligned} \tag{1}$$

where the units of analysis (i) are EU NUTS2[4] regions and the dependent variable (gdp growth) is computed as average annual growth rate of GDP in purchasing power standard (PPS) in two periods: 2008–2012 (economic crisis) and 2012–2016 (post-crisis).[5] The main variables of interest – whose impact on economic performance is expected to be positive, according to the literature – are the regional growth assets identified in the previous section, namely human capital, accessibility, institutions, radical innovation and market innovation. Additional controls include (Table 3): per capita GDP PPS (gdp pc), to control for the initial level of development; a dummy variable (metro) equal to 1 if the NUTS 2 region hosts one or more 'metropolitan regions'[6] to capture the role of agglomeration economies;share of employment in agriculture (sh agr) and share of employment in industry (sh ind) as controls on sectoral composition;

Table 3. Description of control variables

Variable	Description	Computation	Year	Source
gdp pc	Wealth	Per capita GDP PPS	2008; 2012	Eurostat
Metro	Metropolitan region	Dummy variable = 1 if the region includes a 'metropolitan region' (see footnote 3), zero otherwise	2011	European Commission
sh agr	Share of employment in agriculture	Employment in agriculture/total emp.	2008; 2012	Cambridge Econometrics
sh ind	Share of employment in industry	Employment in industry/total emp.	2008; 2012	Cambridge Econometrics

Country (D) and time (T) fixed effects, to control for national and time conditions. In order to ensure as much as possible the expected direction of causality, the variables enter the model according to a lagged temporal consequentiality; they are indeed lagged to the beginning of the growth period explained (2008 and 2012).[7] The second, more in depth, econometric model aims instead at highlighting the specific impact of the same resources in LDRs, even where characterized by different development patterns in terms of sectoral composition. The rationale behind this approach is also related to the stages of development theory (Fisher 1933; Hoover 1948; Hoover and Fisher 1949) that sees regional development as a natural sequence of phases within which productive specialization (from lower to higher value -added sectors) changes physiologically over time. This is especially evident for the first phases of development from agricultural to industrial activities and the process is nowadays particularly visible in Central and Eastern European Countries (CEECs), which are going through a deep transformation from lower to higher value-added economic activities.[8] This situation is evident looking at the share of employment in agriculture, which is much higher in the LDRs than in the others. In fact, agriculture in LDRs involves in 2012 13.6% of the working population vs a sheer 3.4% in the other EU regions. The situation is similar for the employment in industry, which involves 20.1% of the workers in LDRs with respect to 14% in the others.[9] Taking into account all the considerations above, the second step of the analysis is carried out according to the following specifications:

$$\begin{aligned}\text{gdp growth}_{i,t,t+4} = {} & \alpha + \beta_1 \text{ldr} + \beta_2 \text{human capital}_{it} \\ & + \beta_3 \text{ldr}{*}\text{human capital}_{it} + \beta_4 \text{accessibility}_{it} \\ & + \beta_5 \text{ldr}{*}\text{accessibility}_{it} + \beta_6 \text{institutions}_{it} \\ & + \beta_7 \text{ldr}^{*}\text{institutions}_{it} + \beta_8 \text{radical inn}_{it} + \beta_9 \text{ldr}^{*}\text{radical inn}_{it} \\ & + \beta_{10} \text{mkt inn}_{it} + \beta_{11} \text{ldr}^{*}\text{mkt inn}_{it} + \beta_{12} \text{gdp pc}_{it} + \beta_{13} \text{metro}_i \\ & + \beta_{14} \text{sh agr}_{it} + \beta_{15} \text{sh ind}_{it} + \beta_{16} D + \beta_{17} T + \varepsilon \end{aligned} \tag{2a}$$

$$\begin{aligned}\text{gdp growth}_{i,t,t+4} = {} & \alpha + \beta_1 \text{ldr_agr} + \beta_2 \text{human capital}_{it} \\ & + \beta_3 \text{ldr_agr}{*}\text{human capital}_{it} + \beta_4 \text{accessibility}_{it} \\ & + \beta_5 \text{ldr_agr}{*}\text{accessibility}_{it} + \beta_6 \text{institutions}_{it} \\ & + \beta_7 \text{ldr_agr}{*}\text{institutions}_{it} + \beta_8 \text{radical inn}_{it} \\ & + \beta_9 \text{ldr_agr}{*}\text{radical inn}_{it} + \beta_{10} \text{mkt inn}_{it} + \beta_{11} \text{ldr_agr}^{*}\text{mkt inn}_{it} \\ & + \beta_{12} \text{gdp pc}_{it} + \beta_{13} \text{metro}_i + \beta_{14} \text{sh agr}_{it} \\ & + \beta_{15} \text{sh ind}_{it} + \beta_{16} D + \beta_{17} T + \varepsilon \end{aligned} \tag{2b}$$

$$\begin{aligned}\text{gdp growth}_{i,t,t+4} = {} & \alpha + \beta_1 \text{ldr_ind} + \beta_2 \text{human capital}_{it} \\ & + \beta_3 \text{ldr_ ind} * \text{human capital}_{it} + \beta_4 \text{accessibility}_{it} \\ & + \beta_5 \text{ldr_ind} * \text{accessibility}_{it} + \beta_6 \text{institutions}_{it} \\ & + \beta_7 \text{ldr_ ind} * \text{institutions}_{it} + \beta_8 \text{radical inn}_{it} \\ & + \beta_9 \text{ldr_ ind} * \text{radical inn}_{it} \\ & + \beta_{10} \text{mkt inn}_{it} + \beta_{11} \text{ldr_ind} * \text{mktinn}_{it} \\ & + \beta_{12} \text{gdp pc}_{it} + \beta_{13} \text{metro}_i + \beta_{14} \text{sh agr}_{it} \\ & + \beta_{15} \text{sh ind}_{it} + \beta_{16} D + \beta_{17} T + \varepsilon\end{aligned} \tag{2c}$$

where ldr (Equation (2a)) is a dummy variable equal to one if the NUTS2 region is less developed (GDP PPS per capita lower than 75% of the EU average). Ldr_agr (Equation (2b)) is a dummy variable equal to one if the NUTS2 region is less developed and the share of its employment in agriculture is higher than the EU average one. Finally, ldr_ind (Equation (2c)) is constructed according to the same logic for LDRs characterized by a share of employment in industry higher than the European one. The other variables included in this second step of the analysis do not change with respect to what explained above.

These last specifications aim at catching the specific effectiveness of the growth assets of interest in LDRs, even where characterized by different stages of development and development patterns in terms of sectoral composition. They do so through the exploitation of interaction terms that allow to see if the individual resource is used effectively in the LDRs of interest with respect to the other regions.

This second step of the analysis is especially meant to shed light on what resources can be specifically useful in different contexts and on the mechanisms behind their effectiveness, which can in fact be related to pure endowment, to efficient use or to both.

In order to carry out the analysis described above, an EU wide database at NUTS2 level for the two periods was built. Table 2 displays in detail the variables used, their meaning, their computation and the reference years, together with the sources.

The econometric analyses provide the basis for simulating the relative importance of the different growth assets in affecting regional disparities. The methodology for the simulation procedures, which are in fact the most innovative contribution of the present work, is explained in the next subsection.

3.2. Growth assets and regional disparities: simulation procedure

In order to develop the most original contribution of the present work and therefore to assess the effect of different growth assets on regional disparities, a simulation procedure is put in place. Starting from the results obtained through the regression analyses presented before, GDP growth is calculated assuming higher endowment of growth assets in LDRs or higher efficiency in their use by LDRs. Based on the new GDP growth, a simulated GDP per capita is calculated for 2016, on which, on its turn, an indicator of spatial imbalances is computed.

Within this paper, among all regional inequality indexes, the Theil index is applied. The reason for such a choice is that it has the advantage of being decomposable in two or more layers (Akita 2003; Butkus et al. 2018). This peculiarity is important because it allows us to distinguish between- and within-country disparities, which are characterized by different features and which are differently affected by the distribution of growth assets.

The Theil index has been computed on GDP PPS per head according to the following formula, which also shows the decomposition in two components, adding up to the total:

$$T = \sum_r \frac{Y_r}{Y_{EU}} \ln \frac{Y_r/P_r}{Y_{EU}/P_{EU}} = \sum_C \frac{Y_C}{Y_{EU}} \ln \frac{Y_C/P_C}{Y_{EU}/P_{EU}} + \sum_C \frac{Y_C}{Y_{EU}} \sum_{r \subset C} \frac{Y_r}{Y_C} \ln \frac{Y_r/P_r}{Y_C/P_C} \quad (3)$$

where Y stands for GDP PPS, P for population and the indexes EU, C and r stand, respectively, for European Union, Country and region. As can be clearly seen in Equation (3), the total Theil index (the first element) can be decomposed into a 'between-country' index and a 'within-country' index.

The number of LDRs included in the simulations and their characteristics in terms of sectoral specialization (agricultural and industrial) and geographical localization (CEECs or other countries) are presented in Table 4 for 2008 and 2012. As can be clearly seen, most LDRs – and especially all the industrial LDRs – are in CEECs. Moreover, many LDRs are characterized by a higher share of employment with respect to the EU average in both agriculture and industry. These last categories may change between 2008 and 2012 since they are based on the regional and EU level data for the specific year.[10]

More in detail, two sets of simulations are performed. The first one, based on Equation (1), aims at establishing the relative importance of different resources in affecting spatial imbalances and is carried out according to the following procedure:

- in all the LDRs (see before), the endowment of each growth asset (one at a time) is increased to the mean value (in case the region has already an endowment equal or greater than the mean, its value is not changed);
- the regression coefficients are exploited to estimate the GDP PPS in 2016 with the new values. More in detail, starting from the real GDP PPS in 2008, the regression coefficients are used to obtain an estimated GDP PPS for 2012 and from this last estimated value, the regression coefficients are exploited again to get to the estimated GDP PPS for 2016. The residuals are of course considered;

Table 4. Number of LDRs by productive specialisation and geographical location

	2008	2012
Total LDRs	62	62
Agricultural LDRs	49	48
Industrial LDRs	46	48
Both industrial and agricultural LDRs	38	40
LDRs in CEECs	50	50
Agricultural LDRs in CEECs	39	39
Industrial LDRs in CEECs	46	48
Both industrial and agricultural LDRs in CEECs	35	37

- using such estimated GDP PPS for 2016, a simulated GDP PPS per capita is recomputed and the simulated Theil index calculated;
- finally, the simulated Theil index is compared with the real (benchmark) one.

The second set of simulations performed is meant instead to highlight the peculiar role of the growth assets in LDRs characterized by a different stage of development/sectoral specialization. In line with this perspective, and based on the results of regressions (2b) and (2c), these more specific simulations are carried out according to the following procedure:

- in all the agricultural (industrial) LDRs:
  - for those growth assets with an effective use (the coefficient of the interaction term is positive and significant), the endowment is increased (one at a time, to separate the single effect of each asset) to the mean value;
  - for those growth assets with an inefficient use (the coefficient of the interaction term is negative and significant), the efficiency is increased to that of the more efficient regions;
- the regression coefficients are exploited to estimate the GDP PPS in 2016 with the new values, according to the procedure described above;
- the GDP PPS per capita is recomputed and the Theil index recalculated based on the estimated GDP PPS for 2016;
- finally, the real Theil index (benchmark) is compared with the one obtained through the simulation.

The results derived according to the methodology presented above are reported and discussed in the following sections.

4. Determinants of regional growth: econometric results

The results of the econometric models presented above are displayed in Table 5. Although there is empirical evidence that the inclusion of country dummy variables greatly reduces the importance of spatial correlation (Fingleton 2003), spillovers across regional boundaries might still be a factor. Therefore, tests for spatial autocorrelation in the two periods analysed are also reported at the bottom of the table.

The first column presents the results of Equation (1). As expected, and as predicted by the theory, human capital, accessibility, radical and market innovation show a positive and extremely significant impact on regional growth. On the other hand, in our specification, the quality of institutions, despite the positive coefficient, is not statistically significant in explaining economic growth.

As for the controls, the negative sign associated with per capita GDP suggests that a catching-up process – although slower than before – is taking place, with poorer regions growing relatively faster, according to an overall convergence pattern.[11]

The results of econometric specifications (2a), (2b) and (2c) are instead reported in the second (Equation (2a)), third (Equation (2b)) and fourth (Equation (2c)) columns, focusing, respectively, on all LDRs, on agricultural LDRs and on industrial LDRs. The results show that, overall, LDRs do not seem to behave significantly differently with respect to the other regions, the only difference being in their more efficient use of market

Table 5. Regional growth models: econometric results

Variables	Equation (1)	Equation (2a) (dummy all LDRs)	Equation (2b) (dummy AGRICULTURAL LDRs)	Equation (2c) (dummy INDUSTRIAL LDRs)
Human capital	0.234***	0.230***	0.234**	0.231***
	(3.31)	(3.31)	(3.39)	(3.42)
Accessibility	0.086**	0.083*	0.081*	0.083*
	(1.95)	(1.89)	(1.80)	(1.76)
Quality of institutions	0.045	0.044	0.060	0.094
	(0.60)	(0.48)	(0.71)	(1.18)
Radical innovation	0.096***	0.101***	0.107***	0.104***
	(2.83)	(2.97)	(3.12)	(3.01)
Market innovation	0.459***	0.452***	0.440***	0.432***
	(3.58)	(3.38)	(3.40)	(3.22)
Metropolitan regions	0.022	0.023	0.016	0.019
	(0.65)	(0.67)	(0.48)	(0.54)
GDP per capita	−0.441***	−0.442***	−0.433***	−0.427***
	(−3.40)	(−3.26)	(−3.31)	(−3.17)
Share of employment in agriculture	−0.039	0.006	0.008	−0.036
	(−0.53)	(0.08)	(0.09)	(−0.47)
Share of employment in industry	−0.010	−0.030	−0.048	−0.033
	(−0.19)	(−0.58)	(−0.90)	(−0.62)
LDRs of interest		−0.063	0.112	0.158
		(−0.34)	(0.57)	(0.78)
LDRs of interest*human capital		−0.197	−0.432*	−0.370*
		(−0.95)	(−1.82)	(−1.52)
LDRs of interest*accessibility		0.094	0.147**	−0.015
		(1.28)	(2.14)	(−0.17)
LDRs of interest*institutions		−0.013	−0.024	−0.080
		(−0.22)	(−0.40)	(−1.33)
LDRs of interest*radical innovation		−0.034	−0.114**	−0.011
		(−0.53)	(−1.99)	(−0.18)
LDRs of interest*market innovation		0.194**	0.266**	0.189**
		(2.34)	(3.08)	(2.19)
Time fixed effects	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES
No. of observations	515	515	515	515
R^2	0.642	0.650	0.658	0.651

(*Continued*)

Table 5. Continued.

Variables	Equation (1)		Equation (2a) (dummy all LDRs)		Equation (2b) (dummy AGRICULTURAL LDRs)		Equation (2c) (dummy INDUSTRIAL LDRs)	
Tests for spatial autocorrelation	1st period	2nd period	1st period	2nd period	1st period	2nd period	1st period	2nd period
Spatial error								
Moran's *I*	2.161**	0.091	3.290***	0.189	3.380***	0.208	3.311***	0.143
Lagrange multiplier	0.321	3.009*	0.000	2.865*	0.005	2.791*	0.013	2.859*
Robust Lagrange multiplier	0.005	3.410*	0.088	4.582**	0.129	4.427**	0.229	3.627*
Spatial lag								
Lagrange multiplier	1.479	0.523	0.213	0.215	0.211	0.227	0.382	0.391
Robust Lagrange multiplier	1.162	0.924	0.301	1.933	0.335	1.863	0.598	1.159

Dependent variable: GDP PPS average annual growth rate (2008–2012 and 2012–2016).
Standardized coefficients. Robust standard errors. *T*-statistics in parentheses. Significance levels as follows: *10%, **5%, ***1%.
Some spatial autocorrelation in terms of spatial error was detected, using a row-standardized inverse distance matrix (band 0 8). Therefore, for the two periods separately, both OLS and spatial error regressions were run and compared. The results are completely consistent.

innovation. Therefore, we move on distinguishing between differently characterized LDRs in terms of stage of development/sectoral specialization.

As for the agricultural LDRs, the sign and significance of the interaction terms show that these regions seem to be poorly capable of effectively using human capital and radical innovation. This is not particularly surprising, since their development pattern is clearly not based on these specific resources. On the other hand, despite their poor endowment (see Table 1), agricultural LDRs appear particularly effective in the exploitation of accessibility and, especially, market innovation. This last remarkable result is consistent with some previous research (Capello and Cerisola 2019b) and could be related with the ability of these regions to brand local products (e.g. denomination of origin/geographical indication).

Moving to the industrial LDRs (fourth column in Table 5), it is immediately evident how just for a few resources (human capital and market innovation), the results are significantly different from those of the other regions. This could be due in part to the limited level of sectoral disaggregation available in the data. The 'industry' sector in fact includes economic activities that can be extremely different, from traditional manufacturing to high-tech. Nevertheless, although LDRs are in many cases both agricultural and industrial (see Table 4), the difference in the results of the econometric specifications reported in Table 5 proves that, in fact, important distinctions exist between different development patterns. This is indeed the reason why the different models are shown here.

The results obtained from this second step of the econometric analysis lead to think that the final effects on regional economic growth, and consequently on spatial disparities, do depend on both the endowment of resources and their effective use. In addition, these impacts are also determined by the specific characteristics of the areas in terms of different development patterns. This reasoning is investigated in greater depth in the next section, through simulating the impact on regional imbalances of either an increase in the endowment of efficiently used growth assets or an improvement in the effectiveness of resources that are inefficiently exploited.

5. Growth assets and regional disparities: simulation results

The regression outcomes presented in the previous section provide the starting point for the most relevant and original contribution of the present work, i.e. the quantification of the relative importance of different growth assets in the reduction of regional disparities.

In particular, the results of the first simulation refer to an increase in the endowment of key resources up to the mean EU value in LDRs and its effect on the level of wealth (GDP per capita) in the year 2016. Figure 1(a–d) shows the percentage increment applied to 2016 GDP in LDRs through this first set of simulations. As can be seen in the figure through the different scales and as it could be expected from the regression results, acting on human capital and market innovation provides the strongest results, while accessibility and radical innovation play a relatively minor role.

In more detail, the simulation on human capital shows some country effects, with Italy and Portugal being unexpectedly the most advantaged areas together with Bulgaria, while other CEECs – with a level of tertiary education that is already very near or higher with respect to the European average – are less affected. Some sort of country effect is also apparent as far as radical innovation is concerned, but in this case, the Western

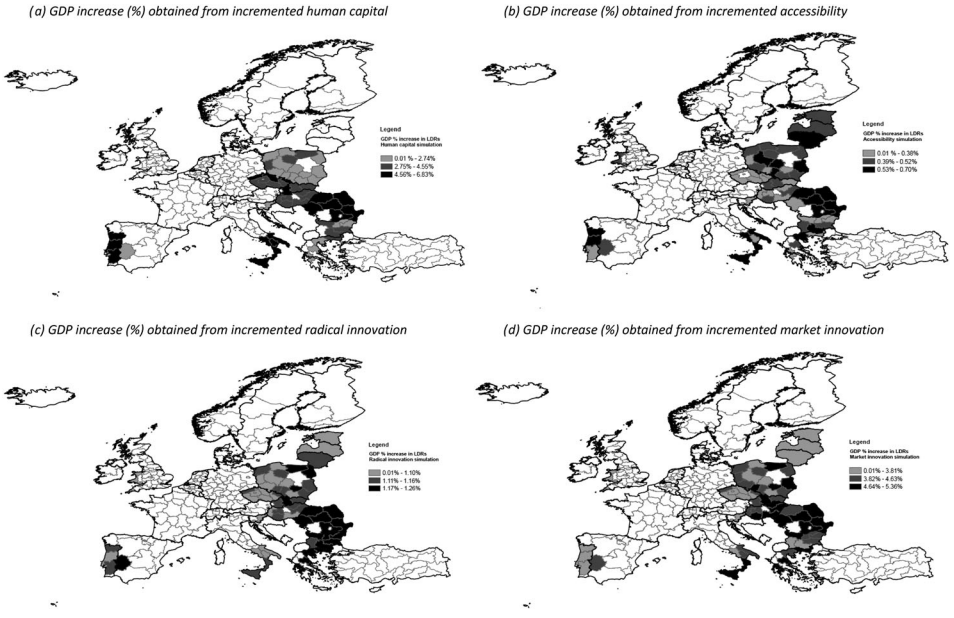


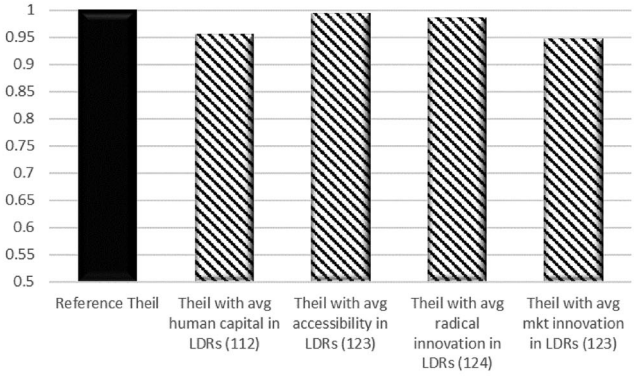
Figure 1 . Simulated 2016 GDP percentage increase in LDRs (from Equation (1)). (a) GDP increase (%) obtained from incremented human capital, (b) GDP increase (%) obtained from incremented accessibility, (c) GDP increase (%) obtained from incremented radical innovation and (d) GDP increase (%) obtained from incremented market innovation.

country (Italy and Portugal) regions gaining less than other countries (e.g. Romania, Bulgaria and Greece). The results involving accessibility and market innovation provide instead a more variegated picture at regional level, suggesting the presence of a high variance in the endowment of such resources within each country.

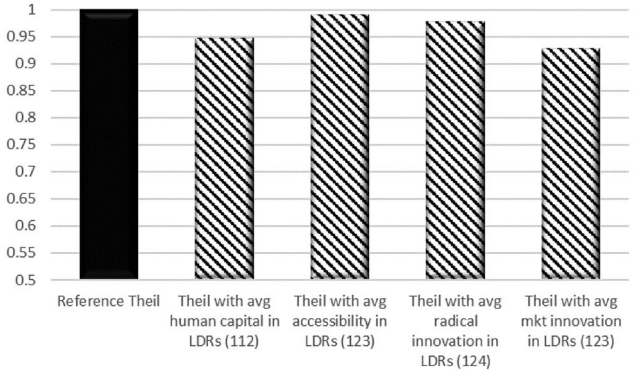
Moving to the simulations on the indicator of regional disparities, the outcome on the total Theil index and on its between – and within-country components is displayed in Figure 2(a–c, respectively). As can be clearly inferred from Figure 2(a), a more equal distribution of resources leads to a decrease in regional disparities, especially when human capital and market innovation are involved. Accessibility and radical innovation, which can also be considered as a proxy for R&D, are instead not as much useful in reducing spatial imbalances. This can take to some considerations on the (possibly limited) effectiveness of a growth based on transport infrastructure (as it is often the case in CEECs) and of a strong push towards R&D everywhere (at least 3% of the GDP to be invested in the R&D sector according to the Europe 2020 Agenda), which does not consider the extremely variegated levels of absorptive capacity characterizing the European regions.

The decomposition of the total Theil index in its between- and within-country components (Figure 2(b,c)) – shows how most of the overall effect on spatial imbalances is in fact due to inter-national disparities. This is mainly related to the inclusion within the LDRs group (thus, the observations modified within the simulation) of almost whole countries in Central and Eastern Europe. Therefore, the simulation takes whole CEECs to a higher level of wealth and this of course decreases between-country inequality.

a – Relevance of each resource on regional disparities in LDRs. Percentage reduction total Theil index - 2016



b – Relevance of each resource on between-country regional disparities in LDRs. Percentage reduction of between-country Theil index - 2016



c – Relevance of each resource on within-country regional disparities in LDRs. Percentage reduction of within-country Theil index - 2016

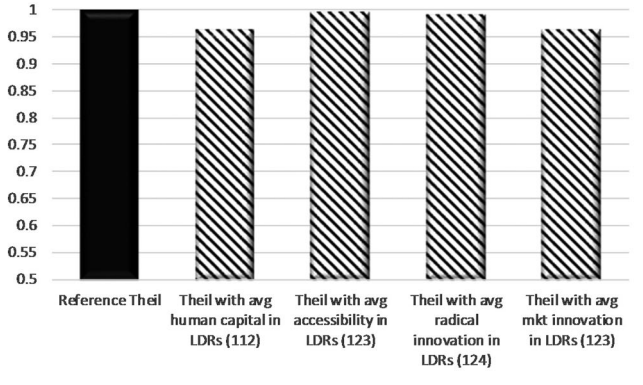


Figure 2. (a) Relevance of each resource on regional disparities in LDRs. Percentage reduction total Theil index – 2016. (b) Relevance of each resource on between-country regional disparities in LDRs. Percentage reduction of between-country Theil index – 2016. (c) Relevance of each resource on within-country regional disparities in LDRs. Percentage reduction of within-country Theil index – 2016. Theil index standardized (real value = 1). Total number of observations modified in the simulation over the two periods are reported in parentheses.

As for intra-national disparities, on the other hand, the effect on the reduction of imbalances is much milder. In this case, however, the specific LDRs involved provide some explanation: raising the endowment of resources in LDRs mainly takes weaker regions nearer to the levels of the more advanced regions in their country, thus lowering within-country disparities in those specific cases. Nevertheless, this happens predominantly in CEECs, so the overall effect at the European level is not particularly pronounced.

This first set of simulations considers that all regions have the same development pattern, in that they require the same resources and use them with the same efficiency. In order to be sure that this assumption does not affect our results, simulations are run taking the specificities of LDRs into account. In particular, the second set of simulations considers that LDRs are not all the same and they are characterized by different development patterns, especially in terms of sectoral composition. In agricultural LDRs, these simulations consider an increase in the endowment of accessibility and market innovation, which are efficiently used resources, and a rise in efficiency in the use of human capital and radical innovation, weakly exploited.

The results of the simulations performed modifying the values related to agricultural LDRs are displayed in Figure 3. As in the previous case, the total Theil index is decomposed in its between- and within-country components and the indexes are standardized (real value equal to 1).

The first remark that can be made from the observation of the results is about the magnitude of the effects, which is in fact much more significant within this second set of simulations than in the previous one, despite the fact that the number of modified values is lower with respect to the first simulation (Figure 2). This tells that it is extremely important to take specificities into account, since they prove to be dramatically relevant.

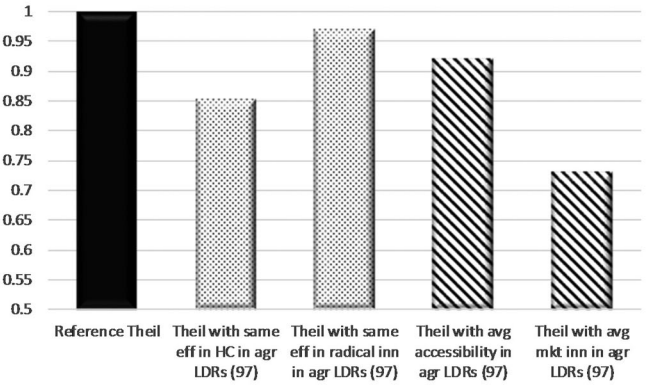
As can be inferred from Figure 3, based on the simulations carried out on agricultural LDRs, the largest advantages in terms of reduction of disparities are obtained through an increase in the endowment of market innovation and through a greater efficiency in the use of human capital.

Agricultural LDRs require, in fact, market innovation to be able to reinforce a pattern of development that is necessarily not based on R&D (the impact of a higher efficiency in the use of radical innovation is relatively limited) but that could be valorized through the capacity to brand local activities/products (market innovation). On the other hand, a more efficient use of the available human capital in agricultural LDRs would improve their economic growth and, consequently, would lead to lower regional imbalances.

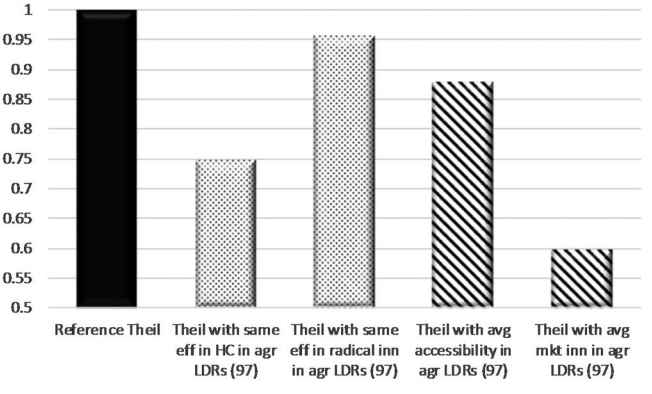
Focusing instead on industrial LDRs (Figure 4), it can be observed how market innovation is in fact not as effective as in the agricultural LDRs, when its endowment is increased. Still, its impact in reducing spatial disparities is higher than the one attained by increasing the efficiency in the use of the available human capital.

A word of caution is here due on the interpretation of the results of the different simulations. They are in fact not completely comparable (especially between Figure 2 and 3 and 4) since a different number of observations is modified according to different rationales. In addition, also the difference between inter-national and intra-national effects should be interpreted carefully since, as explained before, the spatial distribution of regions for which the values were modified changes in the case of between- and within-country disparities and this affects the outcomes.

a – Relevance of each resource on regional disparities in agricultural LDRs. Percentage reduction of total Theil index - 2016



b - Relevance of each resource on between-country regional disparities in agricultural LDRs. Percentage reduction of between-country Theil index - 2016



c - Relevance of each resource on within-country regional disparities in agricultural LDRs. Percentage reduction of within-country Theil index - 2016

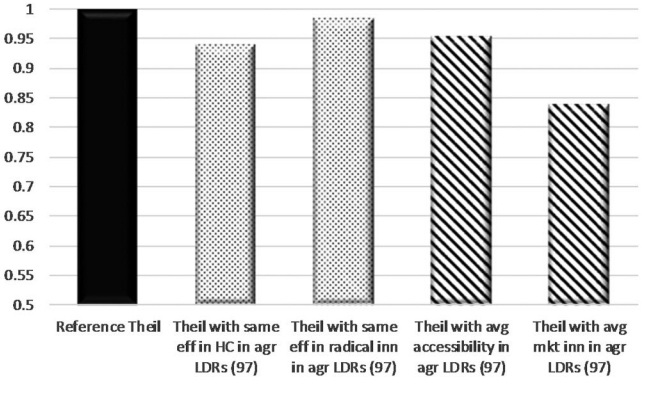
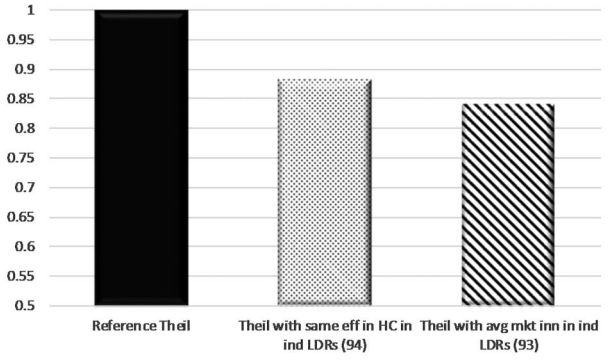
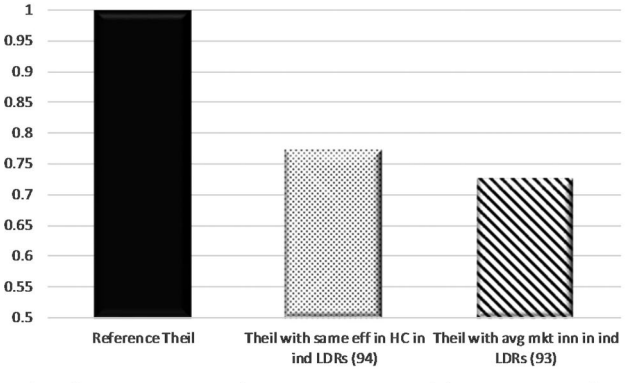


Figure 3. (a) Relevance of each resource on regional disparities in agricultural LDRs. Percentage reduction of total Theil index – 2016. (b)Relevance of each resource on between-country regional disparities in agricultural LDRs. Percentage reduction of between-country Theil index – 2016. (c) Relevance of each resource on within-country regional disparities in agricultural LDRs. Percentage reduction of within-country Theil index – 2016. Theil index standardized (real value = 1). Total number of observations modified in the simulation over the two periods are reported in parentheses.

a - Relevance of each resource on regional disparities in industrial LDRs. Percentage reduction of total Theil index – 2016



b - Relevance of each resource on between-country regional disparities in industrial LDRs. Percentage reduction of between-country Theil index - 2016



c - Relevance of each resource on within-country regional disparities in industrial LDRs. Percentage reduction of within-country Theil index - 2016

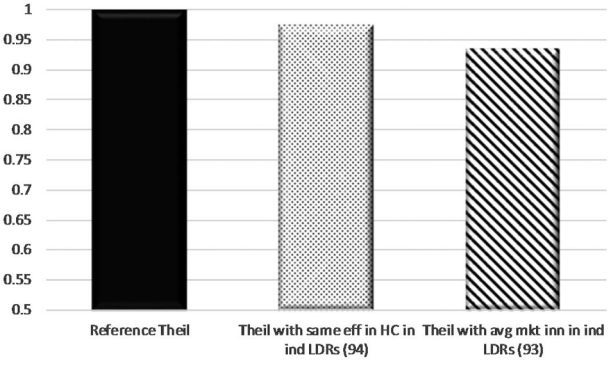


Figure 4. (a) Relevance of each resource on regional disparities in industrial LDRs. Percentage reduction of total Theil index – 2016. (b) Relevance of each resource on between-country regional disparities in industrial LDRs. Percentage reduction of between-country Theil index – 2016. (c) Relevance of each resource on within-country regional disparities in industrial LDRs. Percentage reduction of within-country Theil index – 2016. Theil index standardized (real value = 1). Total number of observations modified in the simulation over the two periods are reported in parentheses.

6. Conclusions

The present study was carried out according to a policy-oriented perspective, in order to raise policymakers' awareness and to support their decisions, providing some useful insights on the most suitable allocation of funding when a disparity reduction strategy is to be pursued.

The work brought to the fore interesting considerations on the relative role of specific resources in the catching-up process of weaker regions and on their subsequent relative importance in the decline of spatial imbalances. The paper provided stimulating and particularly original results, in that it calculated the relative effects of a more evenly spatial distribution of growth assets on regional disparities. These impacts were in fact up to now only logically derived from differentiated effects of local resources between weak and (favoured) strong regions in regional growth models. The intuitive corollary that was derived was that the catching-up of LDRs was influenced by the spatially uneven distribution of resources. A calculation of the specific effects on a disparity index, to our knowledge, was still missing in the literature.

A second important result is that regional imbalances are not exclusively associated with a low endowment of regional resources for growth. In both agricultural and industrial LDRs, resources exist that are inefficiently exploited. In this case, policy interventions should go in the direction of improving the governance and management rather than the endowment of such resources.

The third important type of result was that a higher endowment or a higher efficiency in the use of resources leads to lower regional disparities when the resources reflect the needs of the regional productive structure. The results showed in fact that regional disparities would decrease much more if less developed agricultural regions were endowed with higher capacity to brand their products and activities, than if they had a higher R&D capacity. Public and private R&D investments, in fact, may not succeed if they are not complemented by suitable human capital and supported by favourable context conditions.

Moreover, pure accessibility does not seem to push the catching-up process of LDRs, while human capital shows its importance in both types of LDRs. This result once again reinforces the basic idea of the smart specialization strategy, which calls for a tailored-based design of cohesion policies.

The capacity of the study to identify the individual contribution of different resources in the reduction of regional disparities can be extremely useful. Policymakers can indeed find important suggestions and guidelines in these types of analyses. Studies like this one are in fact able to guide in the decision on where to concentrate investments, i.e. either in supporting job creation, business competitiveness, or in modernizing transport and improving citizens' quality of life, or in reinforcing education, employment, energy, the environment, the single market and R&D.

However, there is of course room for further research aimed at improving and deepening the present work. We plan indeed to go in greater depth into the topic through looking at the role of another important determinant of regional disparities, i.e. the sectoral one. Sectoral composition and efficiency do, in fact, affect productivity growth and therefore contribute to shaping spatial inequalities. In fact, resource availability is only one of the elements behind the possible decrease of regional imbalances. Improving our understanding of different sources of spatial disparities will be more and more important to be able to guide and favour an effective convergence process.

Notes

1. See, among others, Lopez-Bazo et al. (1999), Giannetti (2002), Cappellen et al. (2003), Fischer and Stirböck (2006), Dall'Erba and Le Gallo (2008), Geppert and Stephan (2008), Artelaris, Kallioras, and Petrakos (2010), Del Bo, Florio, and Manzi (2010), Petrakos, Kallioras, and Anagnostou (2011), Smetkowski and Wójcik (2012), Borsi and Metiu (2015), Von Lyncker and Thoennensen (2017), Ayala, Martín-Román, and Vicente (2020), and for a meta-analysis Abreu et al. (2005).
2. We considered Less Developed Regions (LDRs) as those identified for the 2014–2020 programming period, according to the idea that those are the regions on which the European Cohesion Policy mainly focuses at the moment. For a complete list, the reader can refer to the official Commission Implementing Decision, accessed 21 August 2020 at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32014D0099.
3. Source: ESPON, TRACC Project. The indicator was chosen because it combines several modal accessibility indicators (road, rail, inland waterways and air). The *per capita* indicator allows us to have a density measure, capturing the infrastructure supply relative to the potential demand.
4. The 2013 NUTS classification has been considered within the present work.
5. The two periods were chosen considering that 2012 was the first year of recovery from the crisis at the European level (Capello and Caragliu 2020).
6. Metropolitan regions are NUTS3 regions or a combination of NUTS3 regions which represent all agglomerations of at least 250,000 inhabitants (for further details: https://ec.europa.eu/eurostat/web/metropolitan-regions/background, accessed 13 December 2019).
7. See Table 2 for a few exceptions due to data availability.
8. Our intention is not to enter agricultural matters, rather to characterize some LDRs according to their stage of development/specialization pattern. Therefore, the aim is not to suggest agricultural policies, but cohesion policies that work better in regions characterized by an agricultural specialization.
9. ANOVAs showed that in both cases, the statistical difference between the two categories is extremely significant (1%).
10. As we will see later, the number of regions affects the order of magnitude of the simulation results. For this reason, comparison among simulations run on different regional categories must be made with cautiousness.
11. In more detail, after 2008, there was an overall slowing down in the convergence process due to a deceleration in the reduction of *between-country* disparities and to an increase in *within-country* disparities (see OECD 2018; Camagni et al. 2020).

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

A first version of the empirical analysis was supported by Directorate-General for Regional Policy: [Grant Number 2017CE16BAT125].

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